LINUX FORENSICS INVESTIGATION

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System examined: Ubuntu practice VM (hostname: analyst-vm)

Scenario: Suspected compromise (staged practice scenario)

Method: Live + historical examination; suspicious files examined without execution

1. SUMMARY

Three planted artifacts were found and examined: a file in /tmp (update.sh), a note representing a persistence finding (evidence_note.txt), and a recently modified marker (recent_marker.txt). All three were created within a two-minute window (12:16 to 12:18) by the same user, from the same terminal session. No unexpected processes, listening ports, accounts, failed logins, or scheduled tasks were found. Because the artifacts were staged by the investigator, there was no real intrusion. Confidence is high that the artifacts are planted ones, and medium that the rest of the system is clean (limited scope).

2. INVESTIGATOR'S CHECKLIST FINDINGS

PROCESSES (what's running that shouldn't be)

- Reviewed ps aux. Nothing suspicious.
- Processes running as root are expected system services.
- CPU usage was normal for a desktop session.
- No process was running from /tmp.

NETWORK CONNECTIONS (what it's talking to)

- Reviewed ss -tuln. TCP listeners are only on loopback.
- 127.0.0.53:53 and 127.0.0.54:53 (local DNS resolvers).
- 127.0.0.1:631 (CUPS printing, local only).
- No service is listening on an external interface over TCP.
- No SSH server is listening.

LOGS (the history / timeline)

- Reviewed the last 10 lines of /var/log/auth.log.
- All activity is the legitimate user.
- No failed logins, no unknown user, no account creation.

ACCOUNTS (who exists / who can sudo)

- /etc/passwd contains one normal login user plus root.
- Standard system accounts were found.
- Only root has UID 0. The sudo group contains only the analyst's own account.

SCHEDULED TASKS (persistence)

- User crontab (crontab -l): no crontab for the analyst's account.
- System (/etc/cron.d): only "anacron" and "e2scrub_all", which are standard Ubuntu maintenance entries.

FILES (suspicious files)

/tmp/update.sh

- Type (file): ASCII text, not an executable or script binary.
- Properties (ls -l): -rw-rw-r--, owner analyst, 37 bytes, modified Oct 21 12:16. No execute permission.
- Content (cat): fake malicious placeholder, harmless.

Recently modified (find -mtime -1)

- /tmp: update.sh; the rest are root-owned system entries.
- Home: evidence_note.txt and recent_marker.txt, alongside normal desktop cache and config churn.

3. TIMELINE OF EVENTS

Time	Event
12:11:53	sudo apt update run by the user
12:13:12	pkexec update-notifier package lock check
12:15	Terminal opened (ptyxis / bash on pts/0)
12:16:15	/tmp/update.sh created (planted dropped file)
12:17:01	Routine hourly root cron session
12:17:45	~/evidence_note.txt created
12:18:08	~/recent_marker.txt created
12:18	ps aux run (live sweep begins)
12:21:44	sudo tail /var/log/auth.log (log review)

The three artifacts were created in a two-minute burst by the same user in the same terminal, which is what a hands-on session looks like. The timestamp is the thread tying them together.

4. VERDICT AND CONFIDENCE

Verdict: NOT COMPROMISED (staged artifacts identified)

Confidence: HIGH that the artifacts are planted ones; MEDIUM that the rest of the system is clean

Evidence-based reasoning

- The /tmp file is non-executable text, owned by my own account, created at 12:16 during my own terminal session.
- The process list, listening ports, accounts, and user and system cron entries show nothing unexpected.
- auth.log shows only my own sudo and routine system activity, with no failed-login pattern.

5. HOW THE ATTACKER GOT IN AND STAYED

Entry: None (no real attacker). In a real case I would expect to see repeated failed logins followed by a success, or an exploited service, in the logs.

Actions: Planted a file in /tmp (a stand-in for a dropped payload) and created marker files.

Persistence: None found. A real actor would typically use a cron job, a new account, or SSH keys added to ~/.ssh/authorized_keys.

6. RECOMMENDATIONS

- Remove the dropped files and any persistence found.
- Remove rogue accounts and reset compromised credentials.
- Find and remove all persistence: check crontab, system services and authorized_keys.
- Apply hardening to prevent recurrence: keep updates current, apply least privilege, use SSH keys, and monitor auth logs.
- Preserve evidence and escalate per incident process before cleaning up.

7. SCOPE AND LIMITATIONS

This was a staged practice scenario on a clean VM. It demonstrated the investigation workflow and how the planted artifacts surface in find, ls, and file output. A real investigation would add: full log review, mapping every listening port to its process, memory and disk imaging, checks of other persistence points, file-hash lookups, and review of other machines on the network.